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SUPERSEDING
Fed. Spec. GG–S–176
December 22, 1953

FEDERAL SPECIFICATION

SCREEN, X-RAY INTENSIFYING, ROENTGENOGRAPHIC, FOR DIAGNOSTIC RADIOGRAPHY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers intensifying screens for use in diagnostic radiography.

1.2 Classification.

1.2.1 *Types and sizes.* — Intensifying screens shall be of the following types and sizes, as specified (see 6.1).

Type I.—High-speed screens.
Type II. — Medium-speed (general purpose) screens.
Type III. — High-resolutions (detail) screens.
Size 1.—5 by 7 inches.
Size 2.—6½ by 8½ inches.
Size 3.—8 by 10 inches.
Size 4.—10 by 12 inches.
Size 5.—11 by 14 inches.
Size 6.—14 by 17 inches.
Size 7.—Other, as specified.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:
L–F–310—Film, Dental Radiographic; Film, Photofluorographic, Medical; Film, Radiographic, Medical.
PPP–B–585—Boxes, Wood, Wirebound.
PPP–B–601—Boxes, Wood, Cleated-Plywood.
PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.
PPP–B–636—Box, Fiberboard.

Federal Standards:
Fed. Std. No. 102—Preservation, Packaging and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

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Military Specifications:

MIL–B–10377—Box, Wood, Cleated, Veneer, Paper Overlaid.

MIL–L–10547 — Liners, Case, Waterproof.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. — The following documents form a part of this specification. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

American Standards Association, Inc.:

Z38.2.5–1946 — Diffuse Transmission Density.

PH3.19–1948 — Radiographic Intensifying Screens.

(Copies may be obtained from the American Standards Association, Inc., 70 East 45th Street, New York 17, New York.)

American Society for Testing Materials:

ASTM Designation: C190–49 — Tensile Strength of Hydraulic Cement Mortars.

(Copies may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pa.)

3. REQUIREMENTS

3.1 Material. — Satisfactory and high-grade materials entirely suitable for the purpose shall be used throughout. Screens shall be coated on a plastic or cardboard base of weight and quality entirely suitable for the purpose. Cardboard base shall be treated with a moisture- and fungus-resistant preparation.

3.2 Description. — Intensifying screens shall consist of a pair of screens; one to be used as a front screen and one to be used as a back screen.

3.3 Construction. — Screens shall be so constructed as to lie essentially flat without warping or curling, and to insure close apposition of intensifying screens and film when placed in a standard cassette.

3.4 Dimensions. — The dimensions, thickness, and squareness of screens shall be in accordance with ASA Standard PH3.19-1948.

3.5 Surface. — The fluorescent surface of the screens shall be of the cleanable type, and the cleanable surface shall be an integral part of the screen. Manufacturers shall include instructions for cleaning with each pair of screens.

3.5.1 *Cleanability.* — The surface of the screens, when tested as specified in 4.4.1, shall be completely clean and free from staining. The radiograph shall show no sign of artefacts attributable to surface staining.

3.6 Resistance to abrasion. — The resistance of the surface of the screen to abrasion and wear shall be such that, when tested as specified in 4.4.2, the surface of the screen shall show no permanent discoloration nor shall a film exposed with the screen show evidence of artefacts attributable to surface damage to the screen.

3.7 Speed factor. — The speed factor of screens covered by this specification shall be defined as the quotient of the speed of the film-screen combination divided by the speed of the film alone.

3.7.1 *Film-screen speed.*—The speed of the film-screen combination is defined as 100 times the reciprocal of the exposure in milliroentgens required to produce a density of 1.0 above base and fog density in a film exposed between the pair of screens under the standard conditions described in 4.4.3.1.

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3.7.2 *Film speed.*—The speed of the film alone is the speed of the film determined in accordance with Federal Specification L–F–310.

3.7.3 *Screen speed.*—The speed factor of screens covered by this specification shall be as follows, when determined as specified in 4.4.3: Type I (high-speed screens), not less than 1.4; type II (medium-speed screens) between 0.70 and 0.85; and type III (high-resolution screens) between 0.20 and 0.30.

3.8 Uniformity of fluorescing surface. — When tested as specified in 4.4.4, the film exposed between the screens shall show no evidence of mottle or uneven density.

3.9 Persistence. — When tested as specified in 4.4.5, the film shall show no evidence of shadows of the test objects.

3.10 Definition.—When tested as specified in 4.4.6, type I (high-speed) screens shall resolve at least half the holes in Area I (0.024 inches); type II (medium-speed) screens shall resolve at least half the holes in Area II (0.0225 inches); and type III (high-resolution) screens shall resolve at least half the holes in Area III (0.021 inches). Resolution shall be determined by viewing the exposed X-ray film in a viewer without magnification.

3.11 Mounting material. — Manufacturer shall include with each package (pair) sufficient mounting material, i.e., tape, glue, gum, etc., to install screens in cassette, including instructions for using the material. Mounting materials shall be free from any constituents which will produce aerial fog or impair the fluorescent emission by X-ray absorption, chemical reaction, or staining of the active layers.

3.12 Marking for identification. — Each pair of screens shall be marked on the front (screen surface) with the name or registered trademark of the manufacturer, speed type, serial or lot number, contract number, and identification as to "front" or "back" screen, whichever is applicable. If front and back screens are interchangeable, no marking as regards front or back identification is necessary. The marking shall be permanently and clearly visible after screens are mounted in the cassette and shall not encroach on the useful area of the screen. When specified, screens shall be marked "High Speed", "Medium Speed" or "Detail", as applicable (see 6.1).

3.13 Instructions. — Instruction sheet for installation of screens shall be furnished with each pair of screens.

3.14 Workmanship. — Workmanship shall be first class throughout. Screens shall be free from imperfections which affect their serviceability or the diagnostic quality of radiographs made therewith.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection. —The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure suppliers and services conform to prescribed requirements.

4.1.1 *Inspection.* — Inspection as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 *Certificates of quality.*—When available, certificates of quality, supplied by the manufacturer of the component or material,

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listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 *For examination.*—Sampling for examination shall be conducted in accordance with Military Standard MIL–STD–105.

4.2.2 *For tests.*—Sampling for tests shall be conducted in accordance with Military Standard MIL–STD–105. (See table I.)

4.2.3 *Components.*—Sufficient samples of the components shall be selected from each lot of material used in the manufacture of the screens to perform tests as required by the applicable subsidiary specifications or by the tests specified herein. When sample size for components covered by the subsidiary specifications is not specified, the sample size shall be as indicated above. No test result shall fall below the minimum or exceed the maximum values specified. The sample unit for testing shall be the quantity necessary to perform all required tests one time each.

4.2.4 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 **Examination.** — The screens shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests.—Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.4.1 *Cleaning test.*—A few drops of the following test solution shall be placed on the surface of the screen and allowed to dry thoroughly. The spotted sample shall be laid in a darkened area for 24 hours and the dried residue cleaned with a tuft of cotton, using the cleaner recommended by the manufacturer of the screen. The screen sample shall then be subjected to X-ray using 80 KVP of suitable exposure to produce unit radiographic density. The radiograph and screen sample shall be inspected and shall meet the requirements specified in 3.5.1.

Test solution

Sodium chloride ----------1.0 gram.
Lamp black ----------0.5 gram.
Distilled water ----------100 ml.
Shake well before using

4.4.2 *Abrasion resistance.*—Test specimens shall be prepared by cutting pieces 1½ inches square from the sample screen. Mask off a ¼-inch edge all around the specimen by fold-

TABLE I.

Test characteristic	Requirement paragraph	Test paragraph	Inspection level	AQL percent defective
Dimensions	3.4	-----	L5	2.5
Cleanability	3.5.1	4.4.1	L5	*
Abrasion resistance	3.6	4.4.2	L1, but not less than 1	*
Speed factor	3.7	4.4.3	L2, but not less than 2	*
Uniformity of coating	3.8	4.4.4	L2, but not less than 2	*
Persistence	3.9	4.4.5	L2, but not less than 2	*
Definition	3.10	4.4.6	L1, but not less than 1	*

* Acceptance number shall be zero for each of the tests.

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ing half-inch masking tape (do not use cellophane tape) over the edge. The prepared specimens shall be placed in a 1,000-ml. porcelain ball mill charged with 1,000 grams of Ottawa sand, ASTM designation C–190, 20 grams of carbon black, and one 2-inch porcelain ball. Not more than 10 specimens shall be placed in the mill at one time. The mill shall be operated at 35 to 40 r.p.m. for 30 minutes. Upon completion of the milling, the screen specimens shall be removed from the mill and the excess carbon removed with a soft dry cloth. The masking tape then shall be removed, care being taken that none of the front or back coating of the screen is removed with the tape. The screen specimens shall be cleaned by the procedure recommended by the manufacturer until no further dirt can be removed. The central portion of the screen specimen shall be compared with the protected border. There shall be no discoloration of the central area. A radiograph of the screen shall be made as described in 4.4.4. The film shall be examined in the central area and shall meet the requirements of 3.6.

4.4.3 *Speed factor.*

4.4.3.1 *Film-screen speed.*—A cassette shall be properly loaded with the screens under test and a strip of roentgenographic film conforming to the requirements of Federal Specification L–F–310 for type III, class B film. A graduated series of exposures shall be given using roentgen radiation having a quality equivalent to a half-value layer of 4 mm. of aluminum. The exposure for each step shall be 0.10-second duration, and the intensities shall be such as to result in a series of densities in the processed film ranging from approximately 0.6 to 1.6 above base and fog densities in not less than six steps. Each exposure shall be suitably measured in milliroentgens. The temperature of the screen and film at the time of exposure shall be 77° ± 2° F. Following exposure, the film shall be processed according to Federal Specification L–F–310, washed, and dried. Read the densities of the exposed and unexposed portions of the film on a suitable densitometer calibrated to read diffuse transmission density in accordance with ASA Standard Z38.2.5–1946. Plot density against the logarithm of the exposure in milliroentgens. From this plot, determine the exposure required to produce a density of 1.0 above the base and fog density. The speed of the film-screen combination is determined as 100 times the reciprocal of this exposure.

4.4.3.2 *Film speed.*—The speed of a strip of film of the same emulsion number (preferably cut from the same sheet of film as used in 4.4.3.1) shall be determined according to Federal Specification L–F–310.

4.4.3.3 *Speed factor.* — The speed factor of the pair of screens shall be determined by dividing the film-screen speed determined in 4.4.3.1 by the film speed determined in 4.4.3.2.

4.4.4 *Uniformity of coating.*—A suitable cassette shall be properly loaded with the screens under test and a piece of roentgenographic film conforming to the requirements for type III, class B film of Federal Specification L–F–310. A roentgen exposure at approximately 70 KVP shall be given, sufficient to produce in the processed film a density of from 0.6 to 0.8. Following exposure, the film shall be processed according to Federal Specification L–F–310, washed, dried, and examined for uniform density.

4.4.5 *Persistence.*—A suitable cassette shall be loaded with the screens under test but without film. Prior to test, screens shall be rested for 24 hours. The temperature of the screens during the test shall be 77° ± 2° F. Several metal coins or other roentgen-opaque test objects shall be placed on the front of the cassette and an exposure of 12 seconds' duration made, using roentgen radiation at approximately 85 KVP, 5 milliamperes, and 20-inch target-to-screen distance. The cassette shall then be taken to the darkroom and exactly one minute after the termination of the roentgen exposure, the cassette shall be opened and an unexposed piece of type III, class B film (Federal Specification L–F–310) shall be quickly inserted between the screens

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and the cassette shall be closed. After remaining in the closed cassette for 10 minutes, the film shall be removed and processed according to Federal Specification L–F–310, washed, and dried. Examine the film for evidence of shadows of the test objects.

4.4.6 *Definition.*—Two sheets of cast acrylate (i.e. Plexiglas) with a density of between 1.16 and 1.20 grams per cubic centimeter shall be fabricated as follows: The first sheet shall be approximately 10 inches long by 8 inches wide by 4 inches thick. The 4-inch thickness may be made from sheets no less than one inch thick. The second sheet shall be approximately 10 inches long by 6 inches wide by ¼ inch thick. The second sheet shall be divided into three areas by drilling holes of approximately $\frac{3}{32}$-inch diameter around the perimeter of each area. Holes shall be spaced ¼ inch on centers and each area shall be 4 inches by 2 inches. The three areas shall then be drilled with holes spaced ¼ inch on centers; Area I shall have holes 0.024 inches in diameter (Drill No. 73); Area II shall have holes 0.0225 inches in diameter (Drill No. 74); and Area III shall have holes 0.021 inches in diameter (Drill No. 75). The second sheet (¼ inch thick) shall be attached to the first sheet (4 inches thick) such that a 2-inch by 10-inch area of the first sheet is exposed to face the X-ray tube. This 2-inch by 10-inch area shall be cut down one inch such that the thickness in this area is now 3 inches. In the center of this 2- by 10- by 3-inch area, a 2-inch by 2-inch by 2-inch area shall be cut out. In this cutout area shall be placed 8.5 millimeter of aluminum such that one inch of plastic and 8.5 millimeters of aluminum face the tube side. The cut-down adjacent area of the first sheet (3 inches thick) and cutout area will determine the correctness of the contrast developed in the X-ray film and the quality of the radiation. The fabricated acrylate sheets shall then be placed over a bakelite front cassette with screens to be tested and loaded with a piece of type III, class B X-ray Film (Federal Specification L–F–310). An initial exposure shall be made at approximately 80 PKV with 2.5 millimeters of aluminum filtration and 40 inches tube screen distance. Exposure shall be such as to produce a density above base and fog of between 1.25 and 1.75 in the region of the holes. The respective densities in the cutout area and adjacent area (3 inches thick) shall be equal and at least 0.35 but not more than 0.50 density units above the hole region. If the adjacent area density is greater than the cutout area density by more than 0.05 density units, increase PKV. If cutout area density is greater than adjacent area (3 inches thick) by .05 density units, decrease PKV. If the densities of the cutout area and adjacent area

TABLE II.—*Packaging quantities*

Size	Unit package	Intermediate package	Exterior container
	(*Unit*)	(*Units*)	(*Units*)
1	1	24	144
2	Packaging quantities shall be as specified by procuring activity		
3	1	Not required	48
4	1	Not required	36
5	1	Not required	36
6	1	Not required	24

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are equal but not .35 density units above the hole region, the developed contrast is too low. The developed radiograph shall include all areas for density determinations and shall meet the requirements specified in 3.10.

5. PREPARATION FOR DELIVERY

For civil agency procurement the levels of preservation, packaging and packing used are defined in Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Unit of issue.*—One pair of screens, as specified, constitute one unit of issue.

5.1.2 *Packaging quantities.*—The number of units of issue indicated in table II shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in procurement document.

5.1.2.1 *Packing variation permitted.*—If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.*

5.1.3.1.1 *Screen.* — Each screen shall be wrapped in tissue or parchment paper, or packaged in an individual envelope fabricated from tissue or parchment paper.

5.1.3.1.2 *Unit container.* — One piece of cardboard or parchment paper of approximately the same size as a screen shall be interposed between the two wrapped or enveloped screens. One pair of screens (1 unit) shall be packaged in a one-piece folder, envelope style, fabricated from chipboard or fiberboard not less than 0.033 inch in thickness. When a fiberboard folder is used, it shall be further packaged in a paper envelope of suitable size and design. Closure of unit container shall be adequate to prevent spillage of contents under normal handling.

5.1.3.2 *Intermediate package.*—Intermediate package shall be a carton of appropriate size constructed in accordance with Federal Specification PPP–B–636. Each carton shall be adequately secured to prevent accidental opening.

5.1.4 *Level C.*—Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Exterior container shall be designed for type 2 load, and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 3 use; PPP–B–601, table II, using type III, class 1 plywood; PPP–B–621, table III; or Military Specification MIL–B–10377, table II.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.*—Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—Exterior container shall be designed for type 2 load, and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 2 use; PPP–B–621, table I; or PPP–B–636,

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type I, class 1, fabricated from not less than 275-pound test fiberboard.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchases order.

5.3 Marking.

5.3.1 *Civil agencies.*

5.3.1.1 *Interior packages and exterior containers.*—In addition to any marking specified in the contract or order interior packages and shipping containers shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military.*

5.3.2.1 *Interior packages.*—Each unit package and each intermediate package shall be marked in accordance with section 5 of Military Standard MIL–STD–129. Marking shall include the legend: "STORE IN A COOL PLACE (50° – 80°F.)"

5.3.2.2 *Exterior container.*—Exterior container shall be marked in accordance with Military Standard MIL–STD–129. Marking shall include the legend: "STORE IN A COOL PLACE (50° – 80°F.)"

6. NOTES

6.1 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type and size required (see 1.2).
(c) Speed designation when required (see 3.12).
(*d*) Selection of applicable level of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.3 Transportation description. — Transportation description and minimum weights applicable to this item are:

Rail:
Photographers' materials, NOIBN.
Carload minimum weight 20,000 pounds subject to Rule 34 of the Uniform Freight Classification.

Motor:
Photographers' materials, NOI
Minimum weight 20,000 pounds for National Motor Freight Classification.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Army—Army Medical Service
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